



Computer Vision and Image Processing

Lecture 15

Stereo, Multiview Stereo, Light and Perception

- Human beings perceive the **three-dimensional (3D) world** using two eyes. Each eye captures a slightly different image because of the small distance between them. The brain combines these two images and estimates **depth** and **distance** of objects.
- Computer vision attempts to replicate this ability using cameras. A camera captures only a **2D projection** of the real world, losing depth information. Stereo vision and multiview stereo techniques recover this missing depth information by analyzing images captured from multiple viewpoints.

Stereo, Multiview Stereo, Light and Perception

The topics **Stereo Vision, Multiview Stereo, Light, and Perception** are fundamental in:

- 3D reconstruction
- Autonomous vehicles
- Robot navigation
- Augmented Reality (AR)
- Virtual Reality (VR)
- Medical imaging
- Object measurement
- Computational photography

Human Depth Perception

- **Monocular Depth Cues:** These cues require only one eye.

a) Perspective: Objects farther away appear smaller. Example: Railway Track appears to converge at the horizon.

b) Occlusion: If one object blocks another, the blocking object is considered closer. Example: A nearby car blocks part of a distant car. The blocked car is perceived as being farther.

c) Texture Gradient: Texture becomes denser as distance increases. Example: A brick road.

d) Shadows and Lighting: The brain estimates shape and depth using shadows. Example: A ball's shadow on the ground helps us understand its 3D shape and position.

Human Depth Perception

- **Binocular Depth Perception:** Humans have two eyes separated by approximately 6–7 cm. Each eye receives a slightly different image. This difference is called: **Binocular Disparity**.
- The shift between positions of corresponding points in the two retinal images provides information. The brain calculates:

$$Depth \propto \frac{1}{Disparity}$$

- Large disparity → object is close
- Small disparity → object is far

Stereo Vision

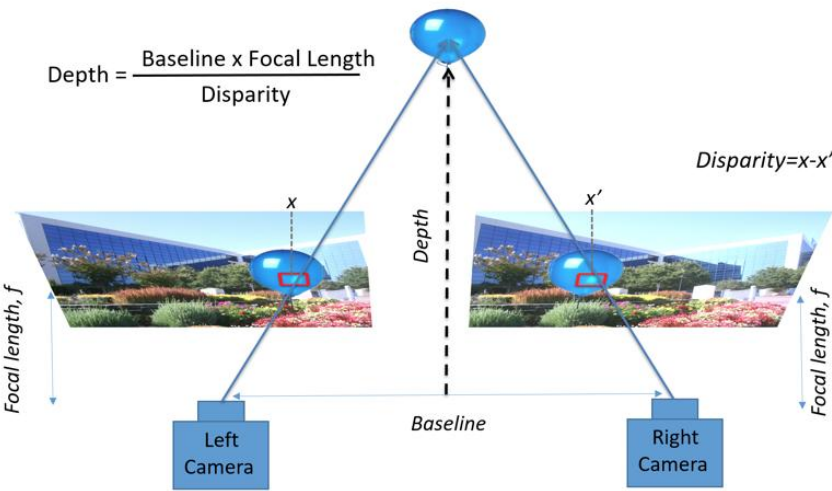
Stereo vision is a computer vision technique that estimates depth from two images captured from different viewpoints.

A typical stereo system consists of:

- Left camera
- Right camera
- Known distance between cameras (baseline)



Disparity Map referenced to Right camera



Right camera

Stereo Matching

- Stereo matching is the process of finding corresponding points between left and right images. The same object point appears at different locations.

$$d = x_L - x_R$$

- **Correspondence Problem:** The major challenge in stereo vision is: "Which pixel in the left image corresponds to which pixel in the right image?"
- Finding correct matches is difficult because of:
 - Similar textures
 - Repeated patterns
 - Occlusion
 - Illumination changes
- image corresponds to which pixel in the right image?"

Stereo Matching

- **Epipolar Geometry:** This is the underlying geometry of stereo vision. If you have a point in 3D space, it projects to a specific pixel in the left image. In the right image, that corresponding pixel is mathematically constrained to lie along a specific 1D line called the **epipolar line**. This reduces the search for matching pixels from a 2D plane to a 1D line, drastically improving computational efficiency.
- **Image Rectification:** Before searching for matches, images are mathematically transformed (rectified) so that the epipolar lines are perfectly horizontal. This means a pixel at coordinate (x, y) in the left image will only match with a pixel at (x', y) in the right image.

Stereo Matching

- **Depth Calculation:** Once disparity d is found (where $d = x_L - x_R$), depth Z is calculated using the focal length f and the baseline B (distance between camera centers) using the formula:

$$Z = \frac{f \cdot B}{d}$$

- This equation shows that depth is inversely proportional to disparity: closer objects have larger disparities, while objects at infinity have near-zero disparity.

Multiview Stereo (MVS)

Multiview Stereo extends stereo vision by using multiple images captured from many viewpoints.

Feature	Stereo Vision	Multiview Stereo
Number of images	2	Multiple
Cameras	Two	Many
Depth accuracy	Moderate	High
Reconstruction	Partial	Complete 3D model
Complexity	Low	High
Applications	Robotics, vehicles	3D scanning, reconstruction

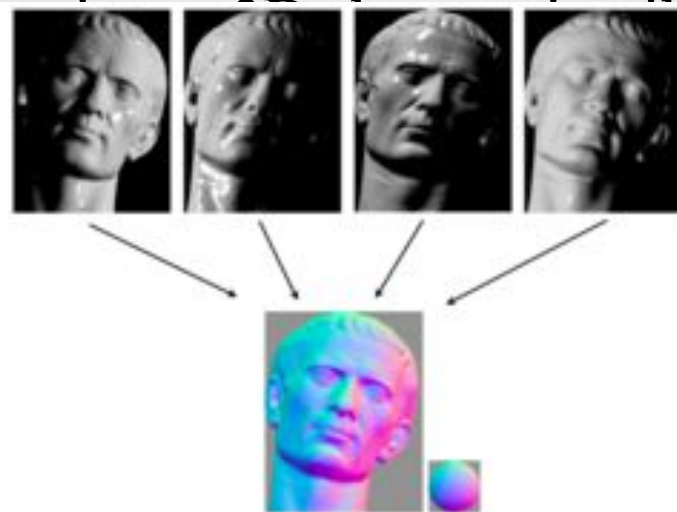
Light

- Light plays an important role because cameras capture reflected light.

- Types of Reflection:

- **Diffuse Reflection:** Light scatters equally in all directions.
 - **Specular Reflection:** Light reflects in one direction.

- Computer vision can use shading to infer shape from illumination. Shape from Shading. Idea : Brightness indicates surface orientation.



Photometric Stereo

- Standard stereo relies on finding matching textures or features. But what if an object is perfectly smooth and featureless, like a porcelain vase? Standard stereo fails. This is where **Photometric Stereo** comes in.
 - Instead of changing the *camera viewpoint*, photometric stereo changes the *direction of light* while keeping the camera completely still.
 - Applications:
 - Face recognition
 - Industrial inspection
 - Medical imaging
- t* while keeping the camera completely still.

Perception in Computer Vision

Computer vision perception means understanding the environment from visual data. A perception system performs : Image Input -> Feature Extraction -> Depth Estimation -> Object Detection -> Scene Understanding

Components of Visual Perception:

- 1. Detection:** Finding objects.
- 2. Recognition:** Identifying objects.
- 3. Localization:** Finding position.
- 4. Scene Understanding:** Understanding relationships.

Applications

Autonomous Vehicles

- Uses:
 - Stereo cameras
 - LiDAR
 - Deep learning
- Tasks:
 - Lane detection
 - Depth estimation
 - Collision avoidance

Applications

- **Robotics**

- Navigate
- Avoid obstacles
- Map environments

- **AR/VR**

- 3D environment reconstruction
- Camera tracking

- **Medical Imaging**

- 3D organ reconstruction

Surgical navigation



The End